Predicting Restaurant Tips

1. Business Objective

The objective of this project was to utilize historical restaurant data to build a mathematical model capable of predicting the tip amount a customer will leave. By understanding which variables most heavily influence gratuities, the restaurant can better forecast staff income and understand customer tipping behaviour.

2. Data Preparation and Cleaning

The raw dataset was imported into Excel and subjected to data quality checks.

- **Missing Values:** The dataset was checked for blank cells using the Go to Special -> Blanks function. No missing values were found, ensuring data integrity.
- **Duplicates:** The Remove Duplicates function was applied to ensure no single transaction was counted twice, which would otherwise skew the mathematical model.

3. Feature Identification and Transformation

To prepare the data for statistical analysis, the variables were categorized:

- **Dependent Variable (Target):** tip
- **Independent Variables (Predictors):** total_bill, size, sex, smoker, day, time.

Because the goal was to predict a continuous numerical value (a specific dollar amount), the problem was identified as a Multiple Linear Regression task. Since regression requires numerical inputs, all categorical text variables were encoded into numeric dummy variables using IF statements (e.g., Female = 1, Male = 0; Dinner = 1, Lunch = 0; Smoker = 1, Non-Smoker = 0).

4. Correlation Analysis

Using the CORREL function, each independent variable was tested against the dependent variable (tip) to determine the strength of the relationships:

- **Strongest Drivers:** total_bill (0.67) and party size (0.49) showed the highest positive correlation, proving that how much a customer spends and the size of their group are the primary drivers of the tip amount.
- **Weakest Drivers:** Demographic and timing factors like sex and time of day showed correlations near zero, indicating they have a negligible isolated impact on tip size.

5. Predictive Modelling

Using Excel's Data Analysis ToolPak, a Multiple Linear Regression model was generated using all independent variables. The resulting coefficients formed the following mathematical equation to predict tip amounts:

Predicted Tip = 0.638 + (0.175 * size) + (0.094 * total_bill) + (0.037 * sex_num) - (0.073 * smoker_num) + (0.053 * day_num) - (0.116 * time_num)

This equation was successfully applied to the historical dataset to generate a Predicted_Tip value for every recorded transaction.

6. Model Evaluation

To evaluate the accuracy of the model, the Root Mean Square Error (RMSE) was calculated by comparing the predicted tips against the actual historical tips.

- **Final RMSE: 1.01**

7.Conclusion & Business Insight: An RMSE of 1.01 indicates that, on average, the regression model's tip predictions are accurate within approximately $1.01 of the actual tip left by the customer. Given the natural unpredictability of customer tipping, this low error rate indicates a very strong and reliable predictive model. The restaurant management can confidently use total expected sales and party sizes as their primary metrics for forecasting tip-based income.